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11.0 INTRODUCTION

Wired and wireless network is by enterprises and end users for accessing information and for performing the tasks. Network enables users to access the required information faster and scale the information access to millions of users.

Mobile device is now all pervasive impacting the daily lives of billions of users. Mobile devices communicate with Internet and Intranet networks for accessing information and for user collaboration. Mobile devices also heavily use public cloud for storage, computing and for secured database.

In this unit we discuss various features of wireless enterprise networks, virtual networks and mobile cloud networks.

11.1 OBJECTIVES

After going through this unit, you should be able to

- understand key concepts of wireless enterprise networks,
- understand security of wireless enterprise networks,
- understand key concepts of virtual network,
- understand key concepts of mobile cloud network.

11.2 WIRELESS ENTERPRISE NETWORK

Enterprises such as banks, business corporates, manufacturing companies provide wired and wireless access to its data for its employees, partners and customers. Corporate employees access the data through intranet networks whereas users access the relevant data through corporate portals and web or mobile platforms.

The enterprise needs to provide secured access to its data meaning that users need to authenticate to access the private data. Post authentication, users should have suitable permissions to access the relevant content.

Enterprises use wide variety of servers, mainframes, firewalls network devices (such as routers), databases to manage and deploy the applications. The core APIs and services are built on top of the existing applications.

We have depicted a typical enterprise network component in Figure 1

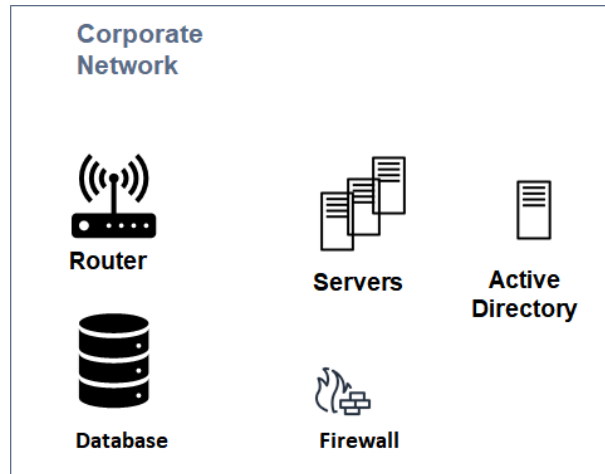


Figure 1 Corporate network components

Wireless enterprise networks provide secured access to the enterprise data and services wirelessly. For instance the employee access the corporate network wirelessly through the mobile apps (such as calendar, corporate email, corporate paging systems) is an example of wireless enterprise network.

Given below are few examples of enterprise applications that normally uses the enterprise wireless network –

- Enterprise Email mobile app
- Enterprise secured browser
- Enterprise calendar mobile app
- Enterprise Intranet portal accessible on mobile device
- Enterprise paging system
- Enterprise user onboarding mobile app

Essentially the mobile apps that access enterprise data and enterprise services use wireless networks.

11.2.1 Features of Wireless Enterprise Network

The wireless enterprise network should be robust to handle various enterprise scenarios. We have depicted various features of wireless enterprise network in Figure 1. The main features of wireless enterprise network can be categorized into five main categories – security, fine-grained access, resilience, scalability and perimeter security.

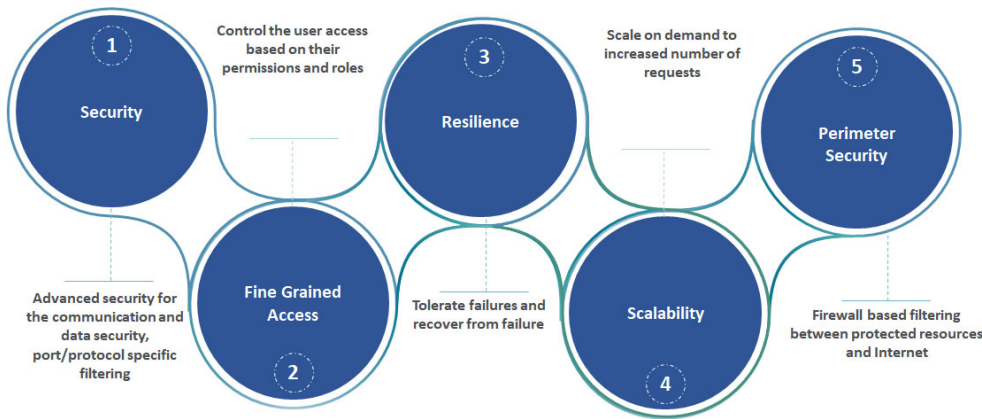


Figure 2 Features of wireless enterprise network

As the enterprise wireless networks provide the access to sensitive corporate information, the core feature of the wireless network is security. The wireless enterprise network should force the authentication for the users to access the sensitive information and allow only the specific protocols and ports. Secondly the wireless network should provide fine grained access based on the user groups and permissions. Thirdly the wireless enterprise network should be resilient and should tolerate and recover from the failures. Fourthly the enterprise wireless network should be scalable to handle a sudden spike in the requests. Lastly the enterprise wireless network should provide the perimeter security through perimeter firewall that filters the traffic between enterprise servers and the Internet.

Enterprises adopt layer-wise and defence-in-depth security model wherein the security policies are implemented at each level. At the enterprise server level, the application allows only the authenticated users. Post authentication, fine grained security is enforced wherein the enterprise authorizes the access to the specific business functionality based on the logged-in user's role and access level. Additionally, the application server logs all the key events (such as password change, login failure and such) to the log files. We also use middleware systems such as Mobile Device Management (MDM) systems that enforce security policies on the end user devices. The security policies enforce the security policies such as access location, device patch level and such.

11.2.2 Components of Wireless Enterprise Server

We have depicted various components of wireless enterprise network in Figure 3.

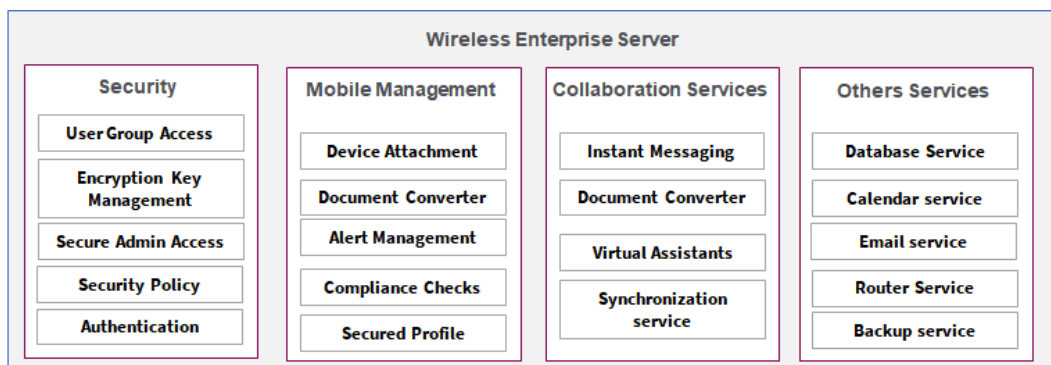


Figure 3 Wireless enterprise server components

We have grouped the components into various categories such as security, mobile management, collaboration services and other services.

In the security category, the wireless enterprise server provide user group and role based access to various services (such as configuration service, administration service and others). The server should also generate the encryption keys that are used for encryption and decryption. The enterprise server also defines various security policy(such as geo-location restriction) and provides the secured administration access. The security policy server pushes the security policies to the mobile devices.

As part of mobile device management (MDM), the enterprise server provides device attachment features wherein the enterprise users can securely register their mobile devices. As part of the device attachment, enterprise users have to provide mobile device details such as mobile OS and mobile identifier. MDM module defines the mobile device compliance rules (such as patch versions) and regularly performs the compliance checks such as mobile OS patch version. MDM also securely open the documents of various formats. MDM also supports other features such as remote wipe

The enterprise server also provides the collaboration service wherein an instant messaging software for users to collaborate among themselves. We could also leverage the connectors that can connect to various third party services such as web services or databases. The synchronization services synchronize the calendar and email on the end mobile devices. Virtual assistants provide chat bots service that users can use to get the information quickly.

The enterprise server provides various other services such as the following –

- Database service to store the user profile and user data
- Email service to manage the user emails
- Push notifications for the mobile device
- Alerts for the enterprise users
- Over the air backup and updates
- Router service to connect the mobile devices to the wireless network
- Backup service for data backup

Check Your Progress 1

1. ____ feature of wireless enterprise network tolerates and handles failures.
2. As part of the ____, enterprise users have to provide mobile device details in Wireless enterprise server.
3. ____ handles the compliance check and remote wiping in Wireless enterprise server.
4. ____ handles the instant messaging in Wireless enterprise server.

11.3 VIRTUAL NETWORK

Networks establish connections among devices, users and domains. Networks also helps us to share information, resources and communicate with each other. Internet is one of the largest networks that allow billions of users and devices to exchange the information.

The physical network uses cables and wires to connect the networks and virtual network extends the features by using software to manage the routing, communication, security and such.

Virtual network is a software-based network that allow different domains or devices to connect to each other over physical network. The communication happens across virtual devices and routers through software-based network. The virtual networks use the same principles of physical network but with reduced operating and maintenance cost.

Enterprises can build virtual network spanning different data centers helping me to build more resilient, fault tolerant and make them more secure. Enterprises can also modify the security rules and network rules easily in the virtual network.

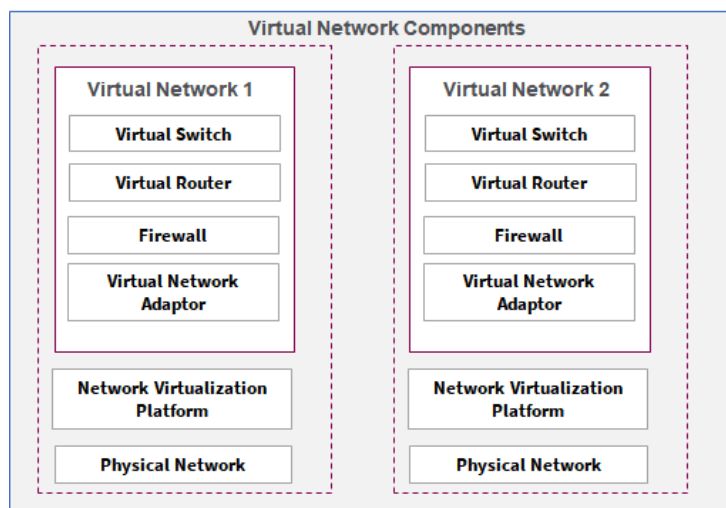
11.3.1 Parameters of Virtual Network

Given below are the parameters of the virtual network –

- Virtual switch – the virtual switch is setup on the host server and is responsible for routing the traffic between virtual network and the underlying physical network.
- Physical network – is the host for the virtual network
- Virtual network adaptor – is the gateway allowing all virtual machines to connect to the network. Using virtual network adaptor, the machines in the Local Areas Network (LAN) can connect to the network.
- Servers – host the server software and part of the virtual network
- Firewalls – monitor the network traffic and detect the threat and anomalous traffic in real time.
- Virtual machines – the machines that connect to the virtual network for exchanging the data

Machines running different operation system (such as Windows, Mac etc.) can do the remote connection using Virtual Network computing (VNC) platform. VNC platform enables the client and server machines to do the remote share of the desktop using the remote frame buffer (RFB) protocol.

We have depicted the parameters of the virtual network in Figure 3.



11.3.2 Types of Virtual Network

Given below are popular types of virtual network –

Virtual Private Network (VPN)

VPNs connect devices over Internet securely (for instance, using encrypted data over IPSec tunnels). VPN allows the home computers or on the go devices to securely connect to the corporate Intranet. There are different types of VPN –

- Site to site VPN wherein we connect two networks using VPN that can be used to connect corporate office to all its branch office networks.
- Client VPN wherein a client device connects to the corporate network securely

Virtual Local Area Network (VLAN)

VLAN segregates the local area network (LAN) into different groups and broadcast domains using software for better security, monitoring and manageability. VLANs provide more control over the network traffic and devices. VLANs also help in securing the data reliably.

11.3.3 Advantages of Virtual Network

Given below are the main advantages of virtual network –

- Virtual networks are the main elements of public cloud platforms that provide hyper scalability and high availability.
- Virtual networks help in centralized, simpler, easier and flexible network management.
- Virtual networks reduce the cost due to the software based segregation.
- Virtual networks provide flexible network routing and configuration.
- Virtual networks provide fine-grained control on the network traffic.
- Virtual networks improve the overall productivity by helping the network administrators automate the setup.

11.4 MOBILE CLOUD NETWORKS

Modern smart phones are part of user's daily life managing various details such as call records, photos, calendar, notes, work related documents, appointments and such. Smart phones are increasingly used to manage huge volume of data. Due to the limited storage capacity of the mobile device, mobile devices leverage cloud computing to offload the storage and computing requirements.

Public cloud platforms are hyper scalar platforms that provide elastic scalability, unlimited storage, high availability and high resilience at lower cost. Public cloud platforms provide highly available network and infrastructure services. Most of the smart phones leverage multiple cloud services to provide seamless user experience.

In this section we shall examine the main cloud services that modern-day smart phones leverage

11.4.1 Cloud Storage Services

Leveraging the cloud storage service is one of the primary use cases for smart phones. As users heavily use smart phones to store photos, media files, the smart phones backup the photos, albums and media files to the cloud storage service for high durability and high availability.

One of the main examples is the cloud drive service offered by Android mobile devices. Users of Android mobile device can backup their media files into Google Drive. Similarly Apple iPhone users can backup their storage on Apple's iCloud services.

In addition to photos and media files, the mobile device also backups contact details, calendar, files, call history, app installation details to the cloud service.

11.4.2 Cloud Compute Services

Few use cases such as deep learning, augmented reality and Artificial intelligence need high compute capacity. For instance, identification of a pattern among the video frames, virtually painting the house with a different color and others need compute intensive machine learning models.

For such use cases, the smart phones send the photos and videos to the cloud services in real time and cloud compute services run the machine learning inferencing jobs and send the results back to the smart phone in real time.

11.4.3 Software as Service (SaaS)

Cloud infrastructure provides various software as service features such as serverless functions that can be used for various use cases such as photo editing service, language translation service, text editor service, messaging service and such. Mobile devices can leverage the SaaS for the use cases.

11.4.4 cLOUD based business applications

Mobile devices can also leverage various business applications such as email application, calendar application, screen sharing application, online meeting application that are hosted on the cloud platforms. The end users can subscribe to those services and use it for personal and business use cases.

11.4.5 Cloud Based Database

Mobile devices can also use the secure database on cloud to manage their sensitive data such as identity documents, contact list, web site passwords and such. The cloud platform uses various encryption algorithms to secure the information on cloud.

☛ Check Your Progress 2

1. Virtual network is a ____ network that allow different domains or devices to connect to each other.
2. _____ is responsible for routing the traffic between virtual network and the underlying physical network.
3. _____ is the gateway allowing all virtual machines to connect to the network.
4. Deep learning use case leverage ____ services of cloud.
5. Email is an example of cloud based _____.

11.6 SUMMARY

In this unit we started by discussing the concept of wireless enterprise network. Applications such as email, calendar, Intranet portal use the enterprise wireless network. The main features of wireless enterprise network can be categorized into five main categories – security (secured communication), fine-grained access (role based access to service), resilience (recover from failure), scalability (on-demand scalability) and perimeter security (firewall based traffic filtering). The enterprise wireless server components can be grouped into various categories such as security, mobile management, collaboration services and other services. Virtual network is a software-based network that allow different domains or devices to connect to each other over physical network. Virtual network provides high scalability, greater flexibility, high availability with lower cost. The main parameters of virtual network are virtual switch, physical network, virtual network adaptor, servers, firewalls and virtual machines. The main types of virtual network are Virtual private network (VPN) and Virtual Local Area Network (VLAN). Mobile devices mainly use the storage, compute, Software as service, business applications and secured database of public cloud platforms.

11.7 SOLUTIONS/ANSWERS

Check Your Progress 1

1. resilience.
2. device attachment.
3. Mobile device management.
4. Collaboration service

Check Your Progress 2

1. software-based.
2. Virtual switch
3. Virtual network adaptor
4. Compute
5. business application

11.8 FURTHER READINGS

References

https://en.wikipedia.org/wiki/Network_virtualization

https://en.wikipedia.org/wiki/Cloud_computing